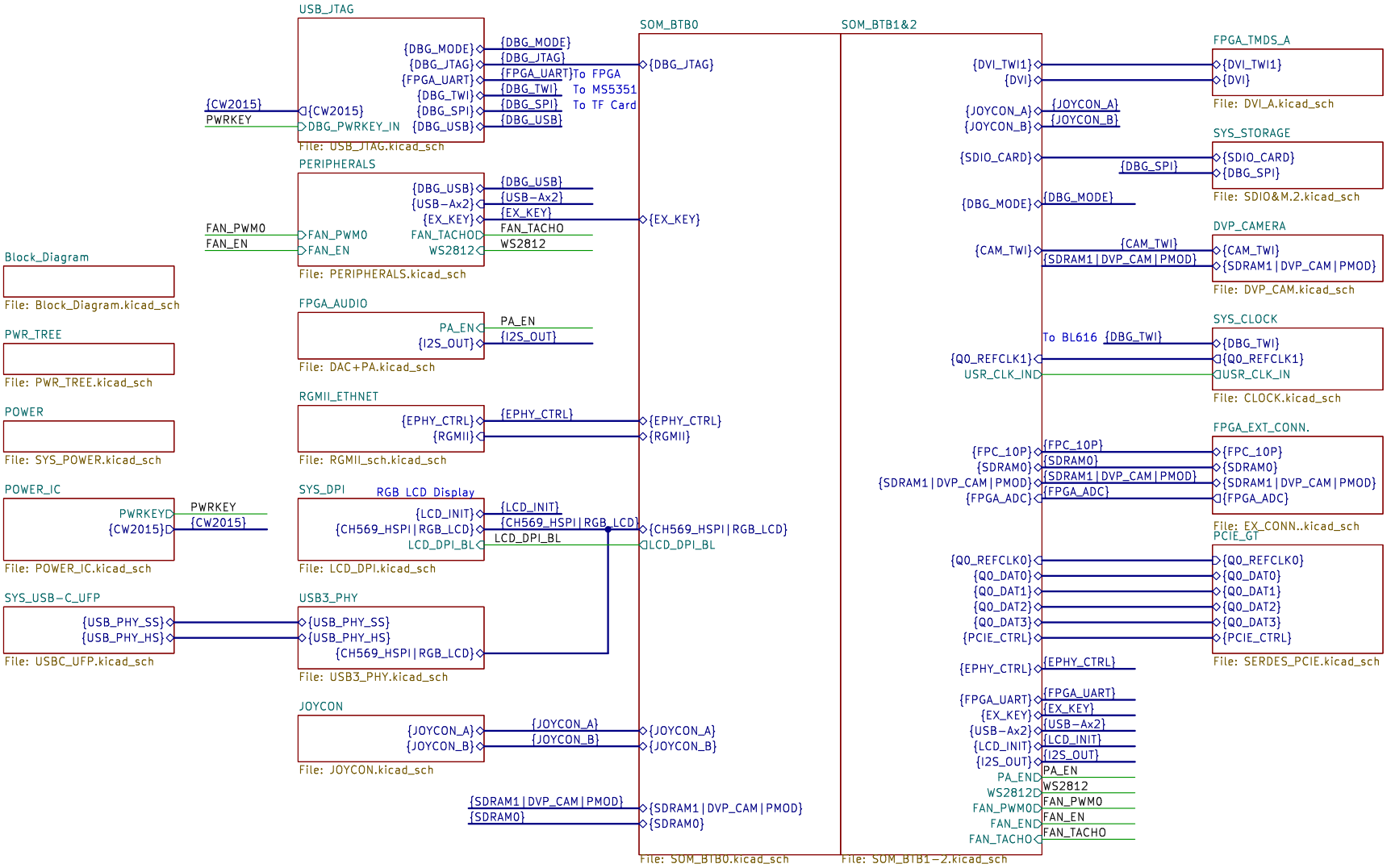
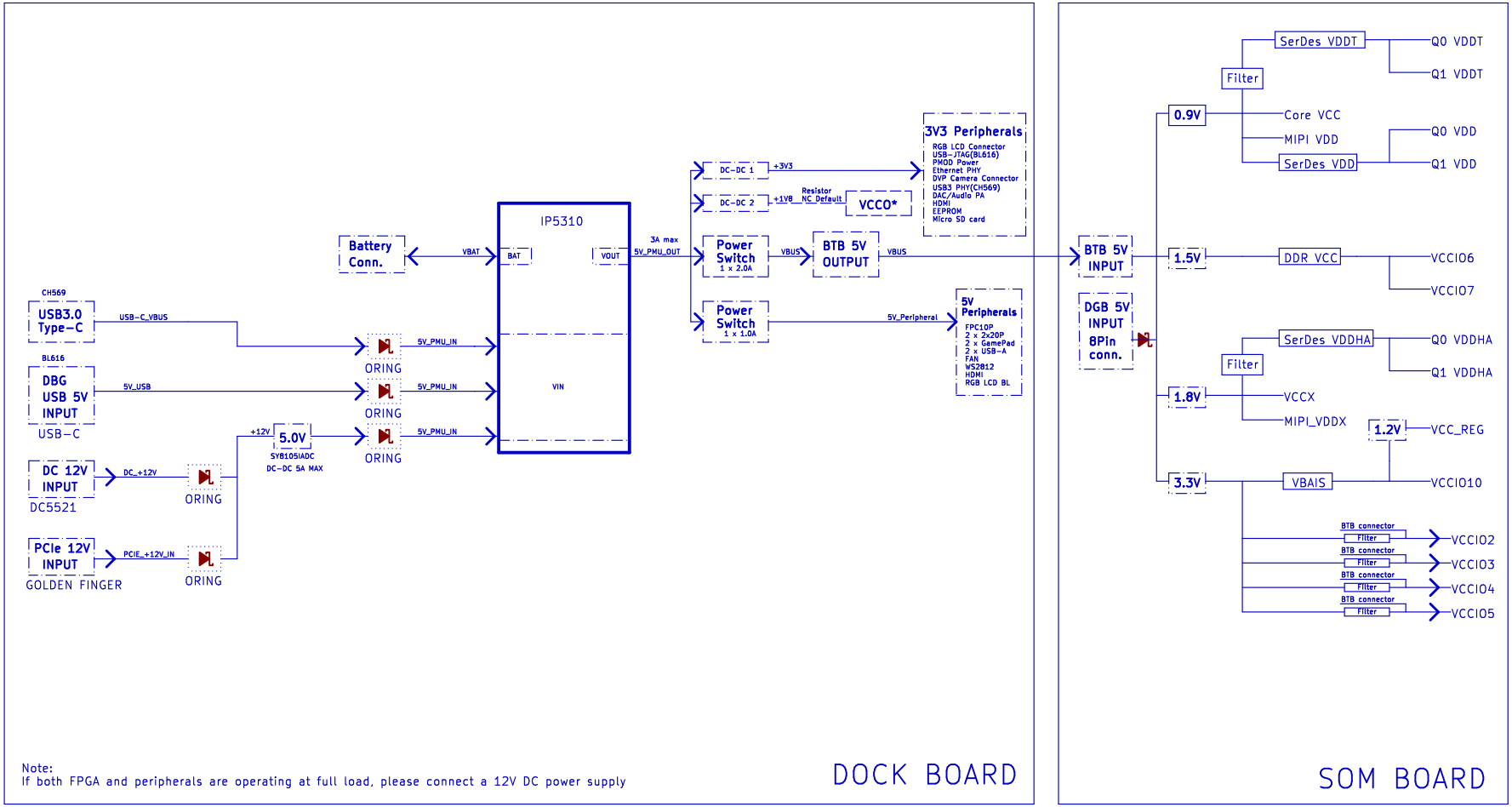

USB_JTAG
{DBG_MODE}
{DBG_JTAG}
{FPGA_UART}
{DBG_TWI}
{DBG_SPI}
{DBG_USB}
{CW2015}
PWRKEY
{CW2015}
DBG_PWRKEY_IN
To FPGA
To MS5351
To TF Card
File: USB_JTAG.kicad_sch
SOM_BTB0
{DBG_JTAG}
SOM_BTB1&2
PERIPHERALS
{DBG_USB}
{USB-Ax2}
{EX_KEY}
FAN_TACHO
WS2812
FAN_PWM0
FAN_EN
FAN_PWM0
FAN_EN
FAN_TACHO
WS2812
File: PERIPHERALS.kicad_sch
{EX_KEY}
Block_Diagram
File: Block_Diagram.kicad_sch
FPGA_AUDIO
PA_EN
{I2S_OUT}
PA_EN
{I2S_OUT}
File: DAC+PA.kicad_sch
PWR_TREE
File: PWR_TREE.kicad_sch
RGMII_ETHNET
{EPHY_CTRL}
{RGMII}
{EPHY_CTRL}
{RGMII}
File: RGMII_sch.kicad_sch
POWER
File: SYS_POWER.kicad_sch
SYS_DPI
RGB LCD Display
{LCD_INIT}
{CH569_HSPI|RGB_LCD}
LCD_DPI_BL
{LCD_INIT}
{CH569_HSPI|RGB_LCD}
LCD_DPI_BL
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LCD_DPI_BL
File: LCD_DPI.kicad_sch
POWER_IC
PWRKEY
{CW2015}
PWRKEY
{CW2015}
File: POWER_IC.kicad_sch
USB3_PHY
{USB_PHY_SS}
{USB_PHY_HS}
{CH569_HSPI|RGB_LCD}
{USB_PHY_SS}
{USB_PHY_HS}
{CH569_HSPI|RGB_LCD}
File: USB3_PHY.kicad_sch
SYS_USB-C_UFP
{USB_PHY_SS}
{USB_PHY_HS}
File: USBC_UFP.kicad_sch
JOYCON
{JOYCON_A}
{JOYCON_B}
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{JOYCON_B}
{JOYCON_A}
{JOYCON_B}
File: JOYCON.kicad_sch
{SDRAM1|DVP_CAM|PMOD}
{SDRAM0}
{SDRAM1|DVP_CAM|PMOD}
{SDRAM0}
File: SOM_BTB0.kicad_sch
{DVI_TWI1}
{DVI}
{JOYCON_A}
{JOYCON_B}
{SDIO_CARD}
{DBG_MODE}
{CAM_TWI}
{SDRAM1|DVP_CAM|PMOD}
To BL616
{DBG_TWI}
{Q0_REFCLK1}
USR_CLK_IN
{FPC_10P}
{SDRAM0}
{SDRAM1|DVP_CAM|PMOD}
{FPGA_ADC}
{Q0_REFCLK0}
{Q0_DAT0}
{Q0_DAT1}
{Q0_DAT2}
{Q0_DAT3}
{PCIE_CTRL}
{EPHY_CTRL}
{FPGA_UART}
{EX_KEY}
{USB-Ax2}
{LCD_INIT}
{I2S_OUT}
PA_END
WS2812D
FAN_PWMOD
FAN_END
FAN_TACHO
{FPC_10P}
{SDRAM0}
{SDRAM1|DVP_CAM|PMOD}
{FPGA_ADC}
{EPHY_CTRL}
{FPGA_UART}
{EX_KEY}
{USB-Ax2}
{LCD_INIT}
{I2S_OUT}
PA_EN
WS2812
FAN_PWM0
FAN_EN
FAN_TACHO
File: SOM_BTB1-2.kicad_sch
FPGA_TMDS_A
{DVI_TWI1}
{DVI}
File: DVI_A.kicad_sch
SYS_STORAGE
{SDIO_CARD}
{DBG_SPI}
File: SDIO&M.2.kicad_sch
DVP_CAMERA
{CAM_TWI}
{SDRAM1|DVP_CAM|PMOD}
File: DVP_CAM.kicad_sch
SYS_CLOCK
{DBG_TWI}
{Q0_REFCLK1}
USR_CLK_IN
File: CLOCK.kicad_sch
FPGA_EXT_CONN.
{FPC_10P}
{SDRAM0}
{SDRAM1|DVP_CAM|PMOD}
{FPGA_ADC}
File: EX_CONN..kicad_sch
PCIE_GT
{Q0_REFCLK0}
{Q0_DAT0}
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{Q0_DAT3}
{PCIE_CTRL}
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File: Tang_Mega_138K_Dock_31004.kicad_sch
Title: Tang_Mega_138K_Dock(NEO)
Size: A4
Date: 2025-03-30
Rev: 1.4
KiCad E.D.A. 8.0.6
Id: 1/21

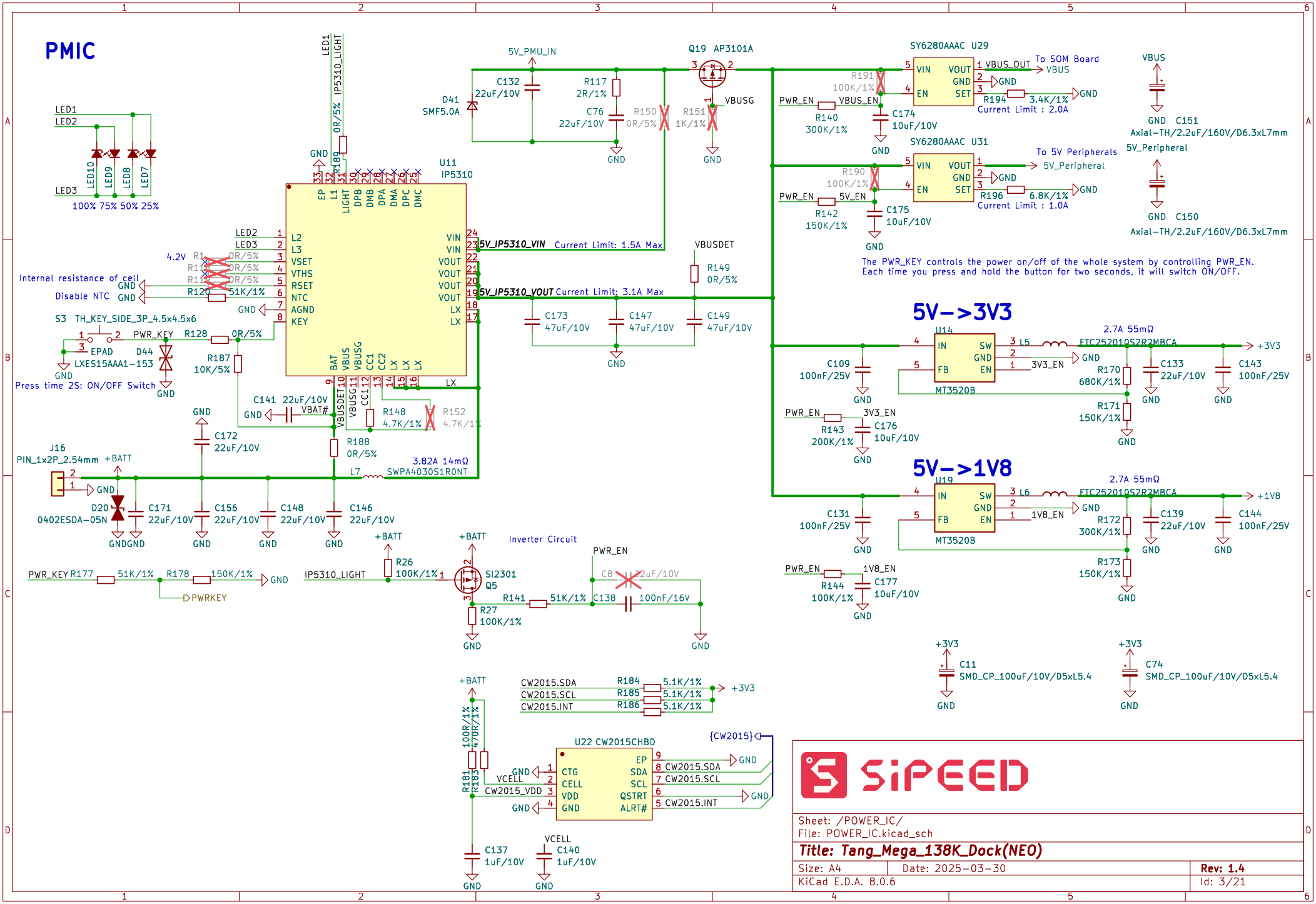


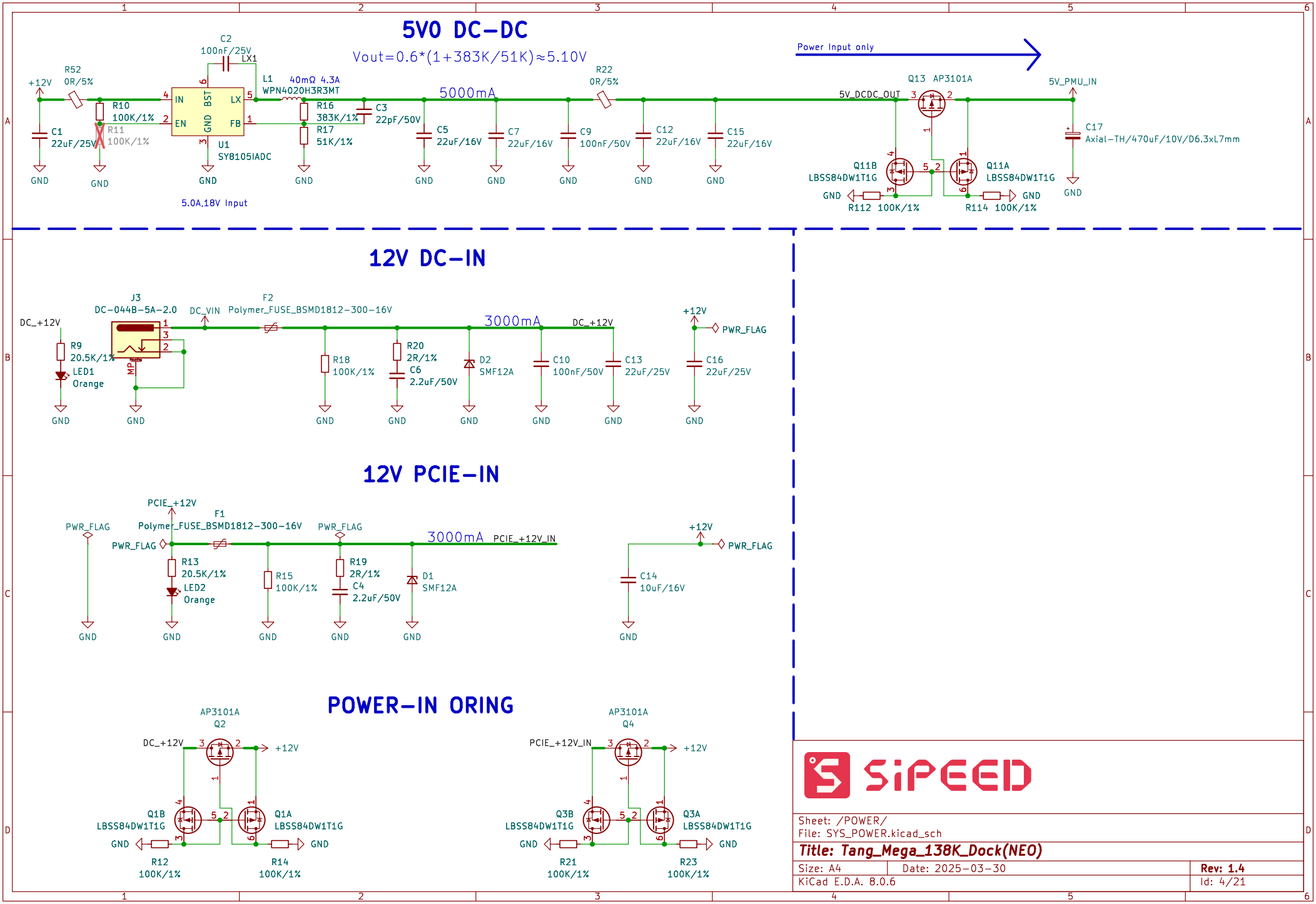
SiPEED

Sheet: /PWR_TREE/
File: PWR_TREE.kicad_sch

Title: Tang_Mega_138K_Dock(NEO)

Size: A4	Date: 2025-03-30	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 2/21


PMIC
LED1
LED2
LED3
LED10
LED9
LED8
LED7
100% 75% 50% 25%
U11
IP5310
5V_PMU_IN
Q19 AP3101A
D41
SMF5.0A
C132
22uF/10V
R117
2R/1%
C76
22uF/10V
R150
0R/5%
R151
1K/1%
VBUSG
SY6280AAAC U29
SY6280AAAC U31
R191
100K/1%
R140
300K/1%
C174
10uF/10V
R194
3.4K/1%
Current Limit : 2.0A
To SOM Board
VBUS
R190
100K/1%
R142
150K/1%
C175
10uF/10V
R196
6.8K/1%
Current Limit : 1.0A
To 5V Peripherals
5V_Peripheral
C151
Axial-TH/2.2uF/160V/D6.3xL7mm
C150
Axial-TH/2.2uF/160V/D6.3xL7mm
5V_IP5310_VIN Current Limit: 1.5A Max
5V_IP5310_VOUT Current Limit: 3.1A Max
VBUSDET
R149
0R/5%
C173
47uF/10V
C147
47uF/10V
C149
47uF/10V
The PWR_KEY controls the power on/off of the whole system by controlling PWR_EN.
Each time you press and hold the button for two seconds, it will switch ON/OFF.
Internal resistance of cell
Disable NTC
S3 TH_KEY_SIDE_3P_4.5x4.5x6
D44
LXES15AAA1-153
Press time 2S: ON/OFF Switch
R128
0R/5%
R187
10K/5%
R120
51K/1%
C141 22uF/10V
VBAT#
R148
4.7K/1%
R152
4.7K/1%
R188
0R/5%
C172
22uF/10V
J16
PIN_1x2P_2.54mm
+BATT
D20
0402ESDA-05N
C171
22uF/10V
C156
22uF/10V
C148
22uF/10V
C146
22uF/10V
L7
SWPA4030S1RONT
3.82A 14mΩ
5V->3V3
U14
MT3520B
2.7A 55mΩ
FTC252010S2R2MBCA
C109
100nF/25V
R143
200K/1%
C176
10uF/10V
R170
680K/1%
R171
150K/1%
C133
22uF/10V
C143
100nF/25V
3V3_EN
+3V3
5V->1V8
U19
MT3520B
C131
100nF/25V
R144
100K/1%
C177
10uF/10V
R172
300K/1%
R173
150K/1%
C139
22uF/10V
C144
100nF/25V
1V8_EN
+1V8
PWR_KEY
R177
51K/1%
R178
150K/1%
PWRKEY
IP5310_LIGHT
R26
100K/1%
Q5
SI2301
Inverter Circuit
R141
51K/1%
C138
100nF/16V
R27
100K/1%
C8
22uF/10V
C11
SMD_CP_100uF/10V/D5xL5.4
C74
SMD_CP_100uF/10V/D5xL5.4
CW2015.SDA
CW2015.SCL
CW2015.INT
R184
R185
R186
5.1K/1%
U22 CW2015CHBD
{CW2015}
R181
100R/1%
R183
47OR/1%
VCELL
CW2015_VDD
C137
1uF/10V
C140
1uF/10V
SiPEED
Sheet: /POWER_IC/
File: POWER_IC.kicad_sch
Title: Tang_Mega_138K_Dock(NEO)
Size: A4
Date: 2025-03-30
Rev: 1.4
KiCad E.D.A. 8.0.6
Id: 3/21



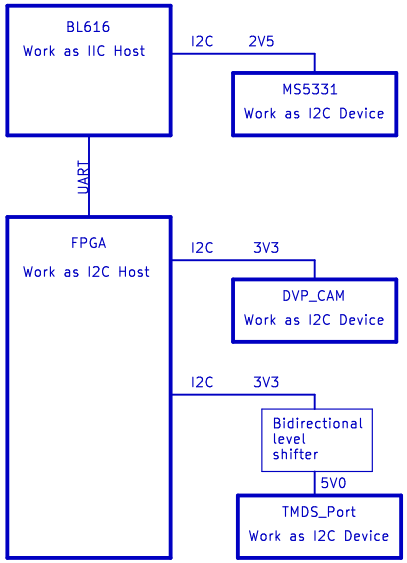
SiPEED

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File: SYS_POWER.kicad_sch

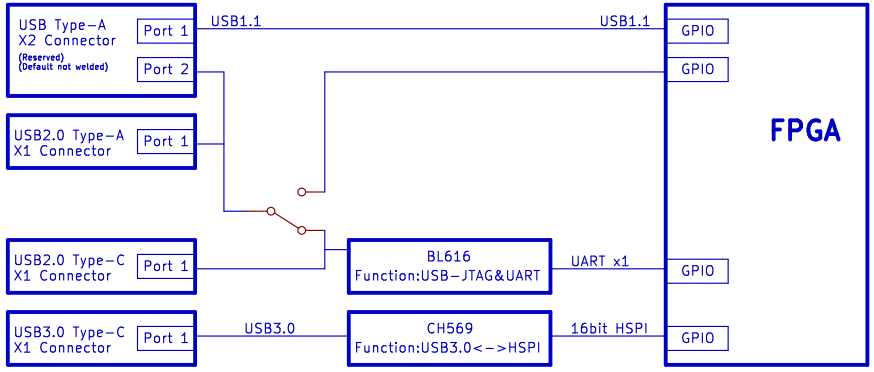
Title: Tang_Mega_138K_Dock(NEO)

Size: A4	Date: 2025-03-30	Rev: 1.4
KiCad E.D.A. 8.0.6		Id: 4/21

IIC connection



USB connection

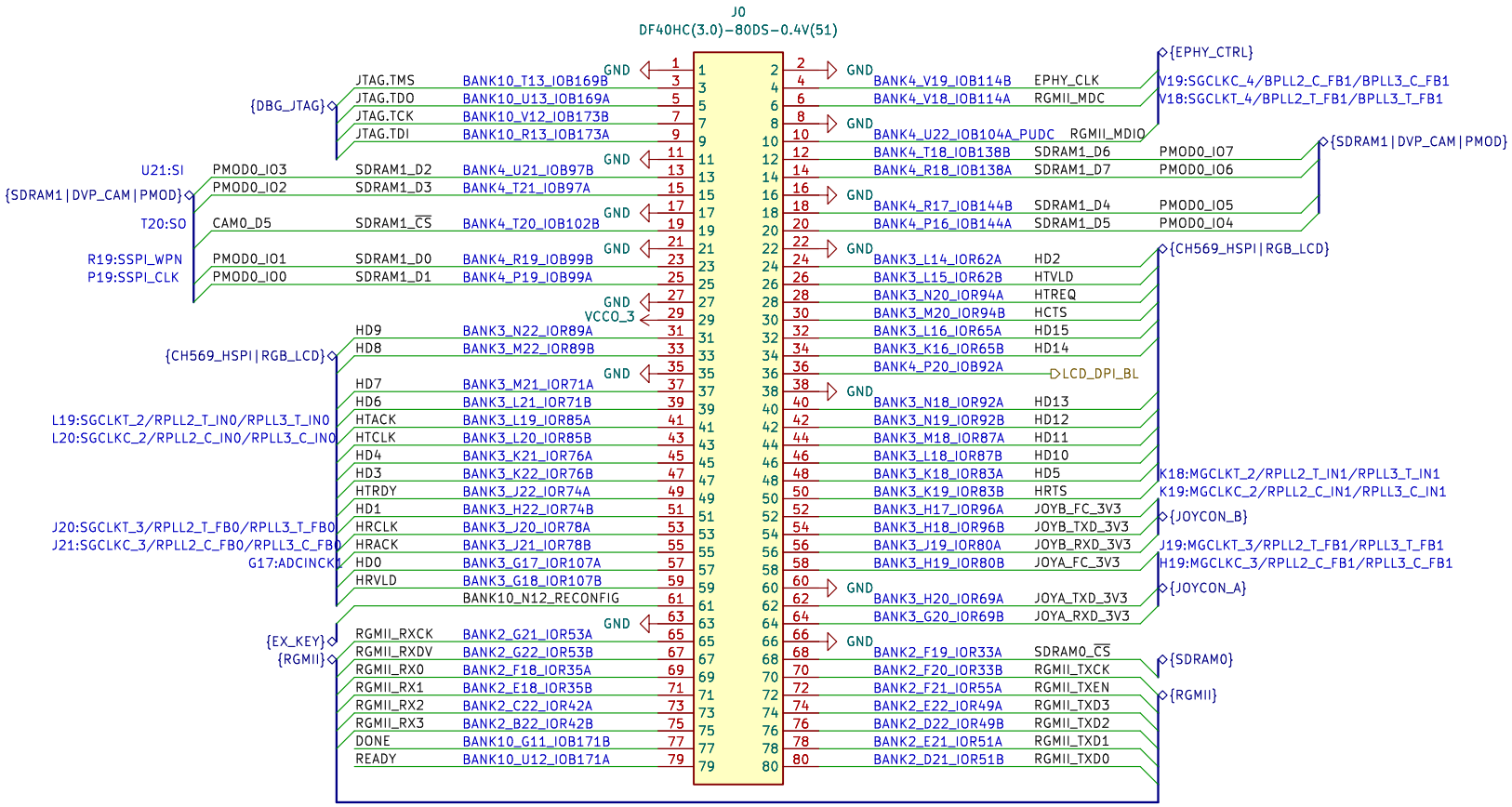


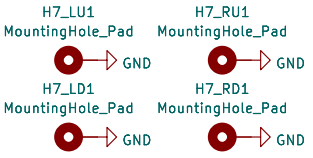
SiPEED

Sheet: /Block_Diagram/
File: Block_Diagram.kicad_sch

Title: Tang_Mega_138K_Dock(NEO)

Size: A4	Date: 2025-03-30	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 5/21







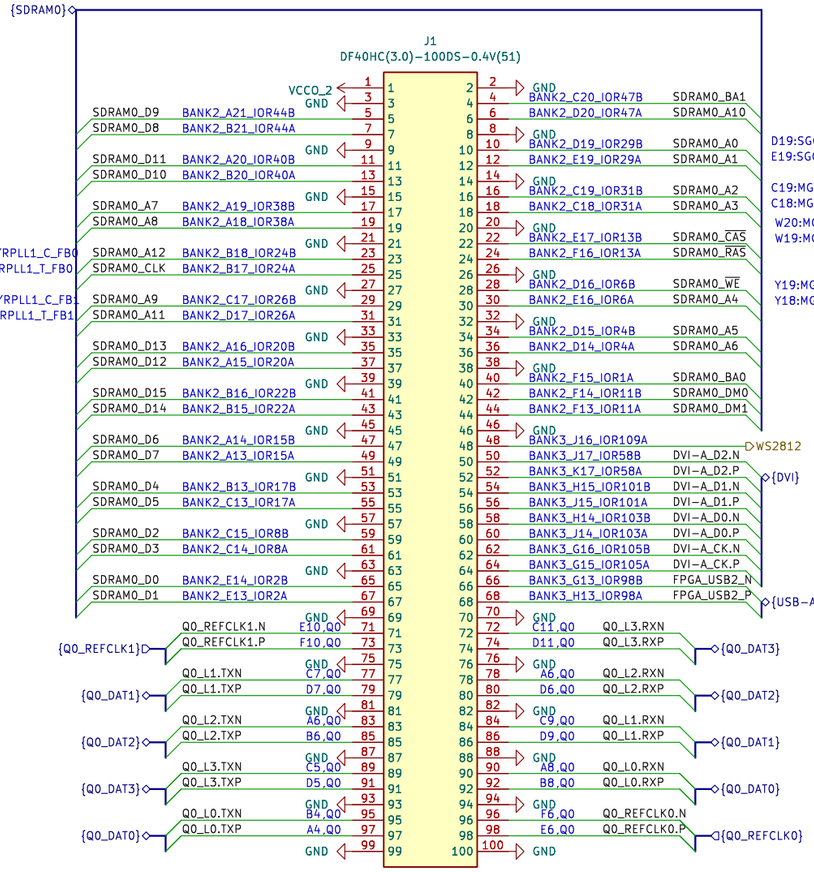


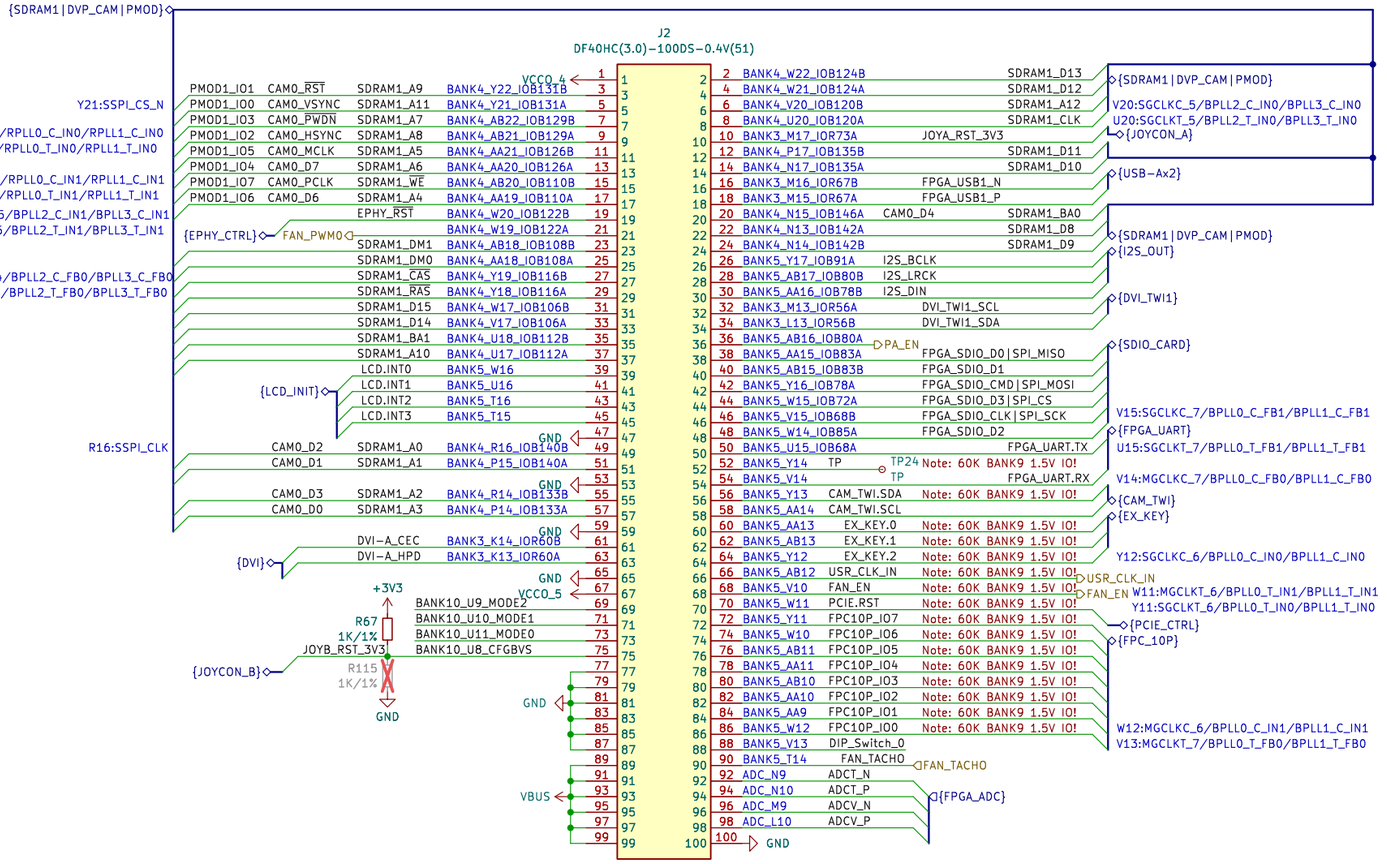
SiPEED

Sheet: /SOM_BTB0/
File: SOM_BTB0.kicad_sch

Title: Tang_Mega_138K_Dock(NEO)

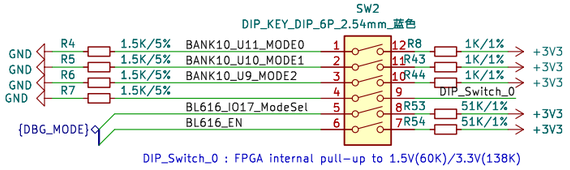
Size: A4	Date: 2025-03-30	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 6/21





VCCO_2 TP20 TP
VCCO_3 TP21 TP
VCCO_4 TP22 TP
VCCO_5 TP23 TP

Switch



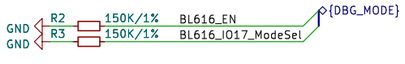
GW5AT-138/GW5AST-138 Configuration Modes

Configuration MODE	MODE[2..0]	Bus Width	Description
JTAG	XXX/101	–	GW5AT & GW5AST series of FPGA products are configured by external Host via JTAG interface
MSPI	001	x1, x2, x4	As a Master, FPGA reads data from external Flash (or other devices) via the SPI interface for configuration
SSPI	010	x1, x4	GW5AT & GW5AST series of FPGA products are configured by external Host via SPI interface
Master SERIAL	000	x1	Before FPGA used as a Slave FPGA reads data from external devices via the DIN interface for configuration
Slave SERIAL	111	x1	GW5AT & GW5AST series of FPGA products are configured by external Host via DIN interface
Master CPU	100	x8, x16, x32	Before FPGA used as a Slave FPGA reads data from external devices via the DBUS interface for configuration
Slave CPU	110	x8, x16, x32	GW5AT & GW5AST series of FPGA products are configured by external Host via DBUS interface

Please refer to GOWIN Manual UG984/UG984E for more details.

BL616 SEL[0..1] SWITCH

NO.	CONFIG	SEL0(EN)	SEL1(MODE)
0	Disable BL616 Use other debugger	0	X
1	JTAG mode	1	1
2	Game mode	1	0



SiPEED

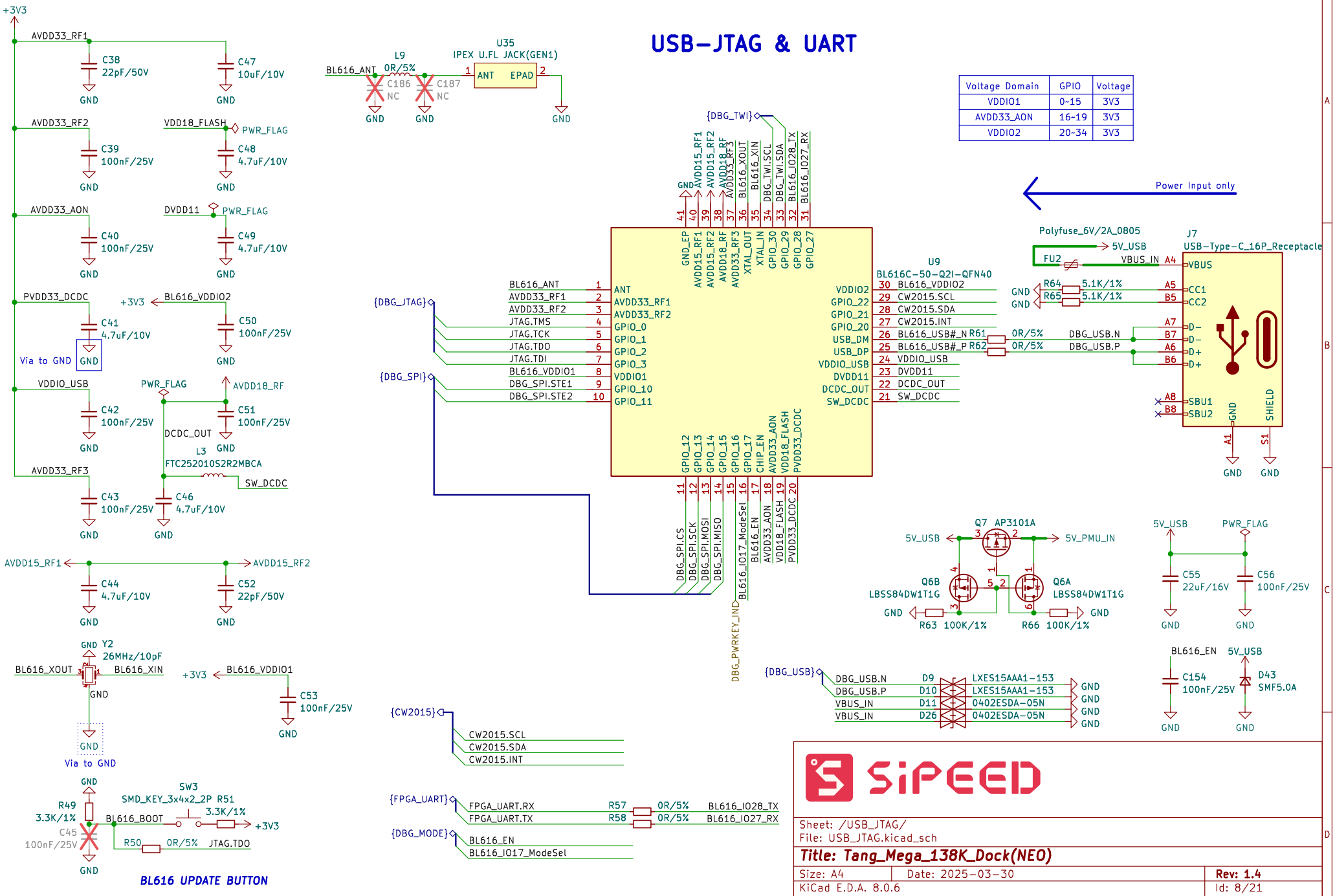
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File: SOM_BTB1-2.kicad_sch

Title: Tang_Mega_138K_Dock(NEO)

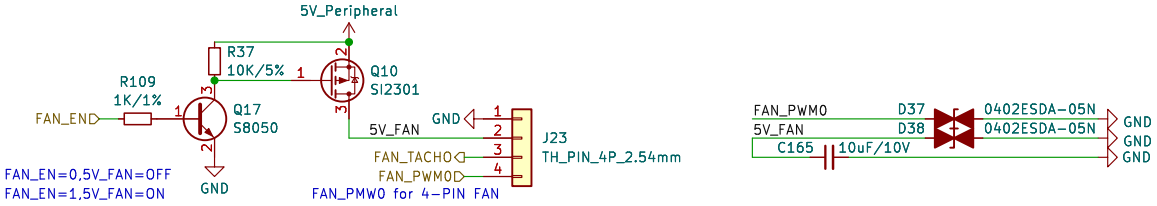
Size: A3	Date: 2025-03-30	Rev: 1.4
KiCad E.D.A. 8.0.6		Id: 7/21

USB-JTAG & UART

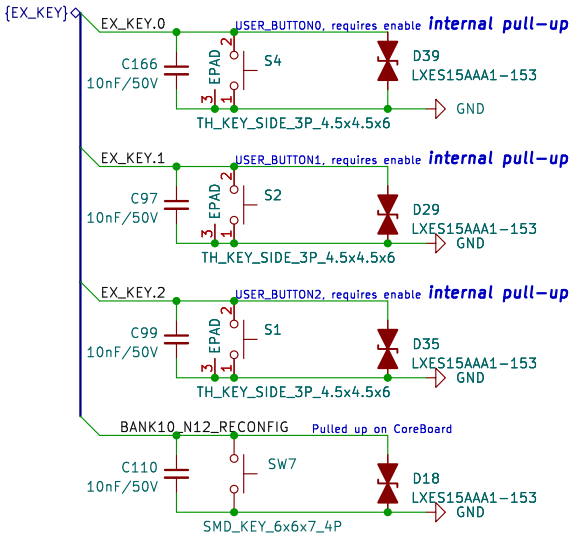
Voltage Domain	GPIO	Voltage
VDDIO1	0-15	3V3
AVDD33_AON	16-19	3V3
VDDIO2	20-34	3V3



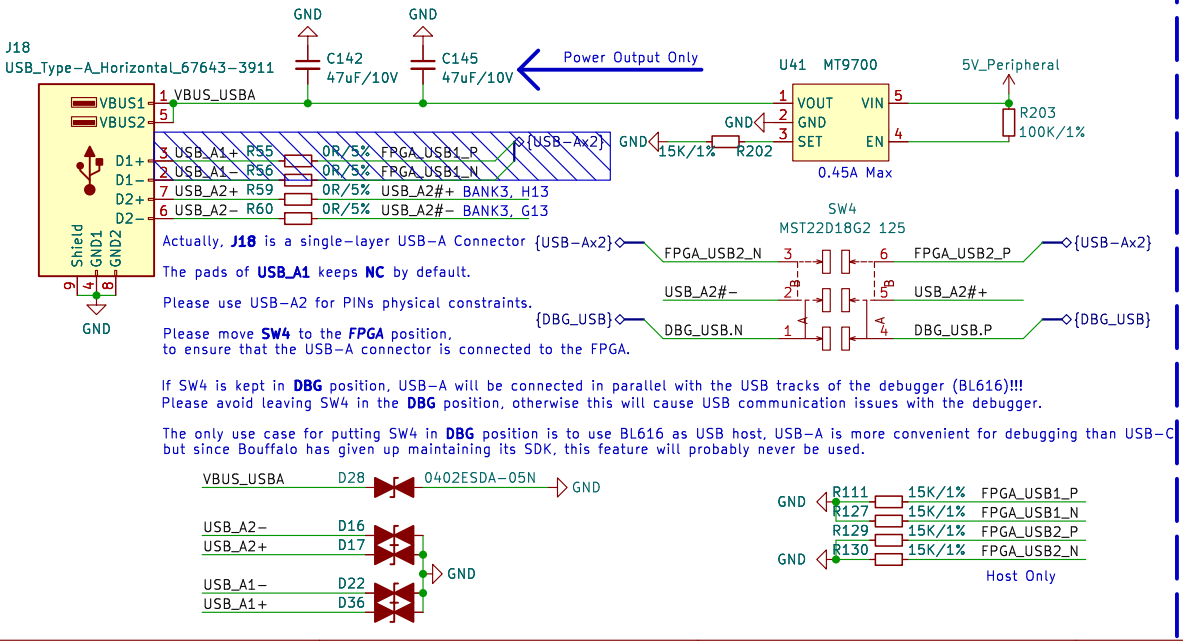
FAN Connector



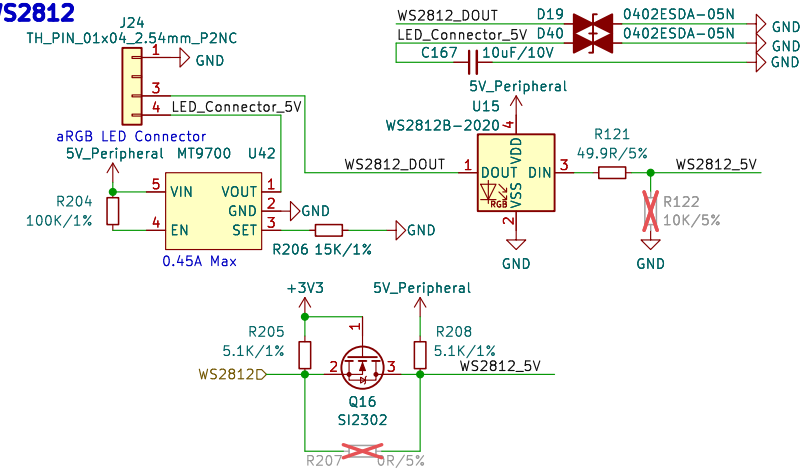
BUTTON



USB_A_x2



WS2812



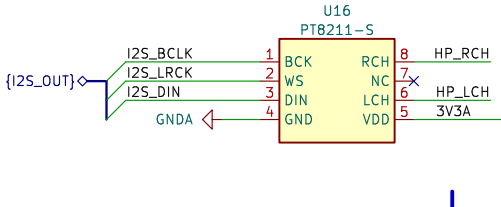
SiPEED

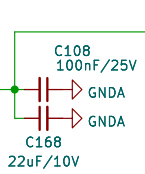
Sheet: /PERIPHERALS/	
File: PERIPHERALS.kicad_sch	
Title: Tang_Mega_138K_Dock(NEO)	
Size: A4	Date: 2025-03-30
KiCad E.D.A. 8.0.6	Id: 9/21

Rev: 1.4

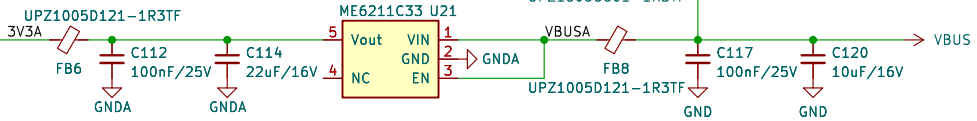
FPGA-Audio

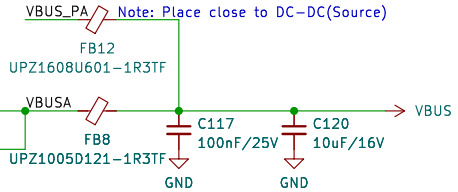
STEREO DAC

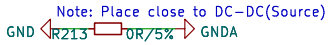




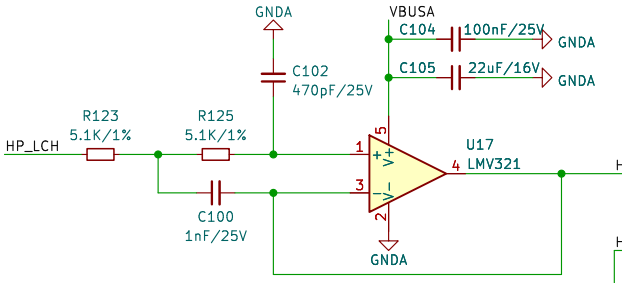
AUDIO POWER

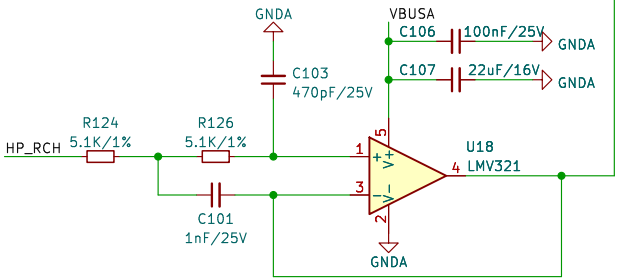


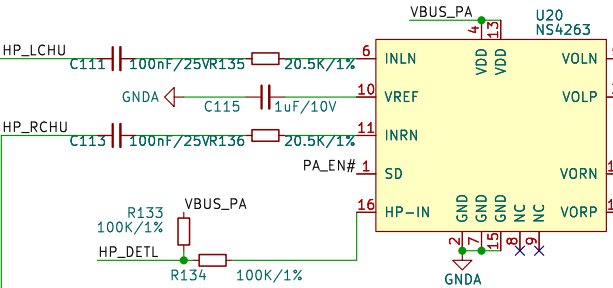


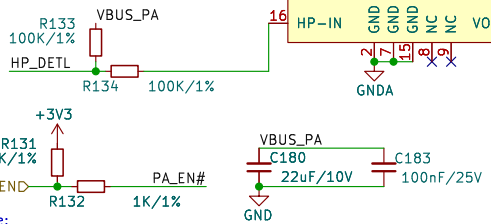


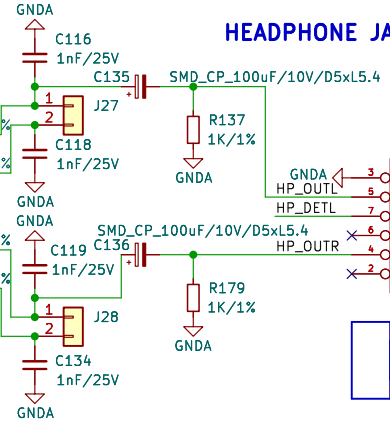
LOW-PASS FILTER



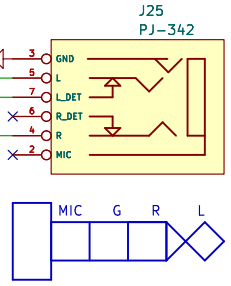








HEADPHONE JACK 3.5MM CTIA



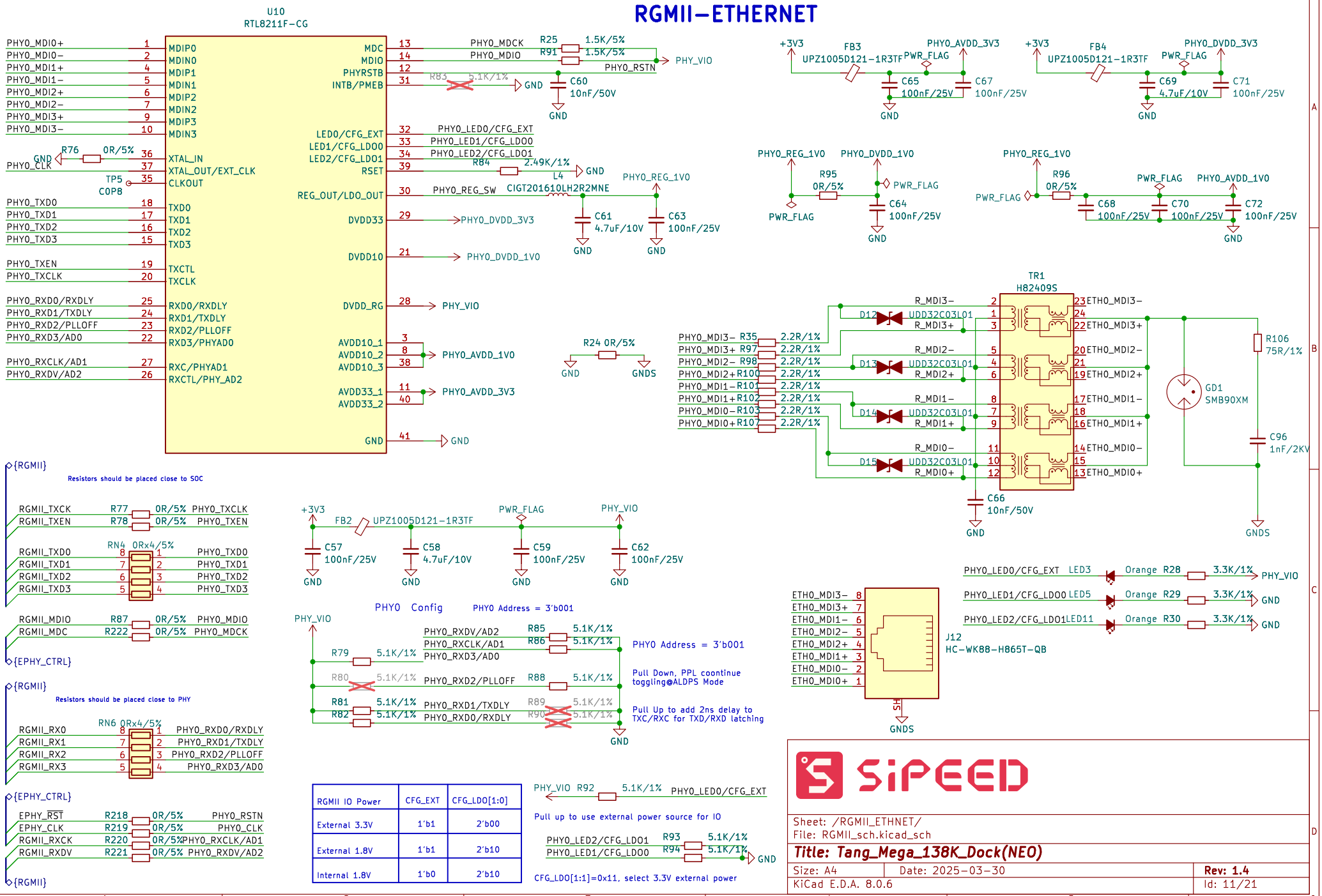
When the jack is unpluged, L/R_DET is connected to L/R.
HP-IN = 0
Speaker enable

When the jack is pluged, L/R_DET is disconnected to L/R.
HP-IN = 1
Headphone enable

SiPEED

Sheet: /FPGA_AUDIO/		
File: DAC+PA.kicad_sch		
Title: Tang_Mega_138K_Dock(NEO)		
Size: A4	Date: 2025-03-30	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 10/21

RGMII—ETHERNET



RGMII IO Power	CFG_EXT	CFG_LDO[1:0]
External 3.3V	1'b1	2'b00
External 1.8V	1'b1	2'b10
Internal 1.8V	1'b0	2'b10

SiPEED

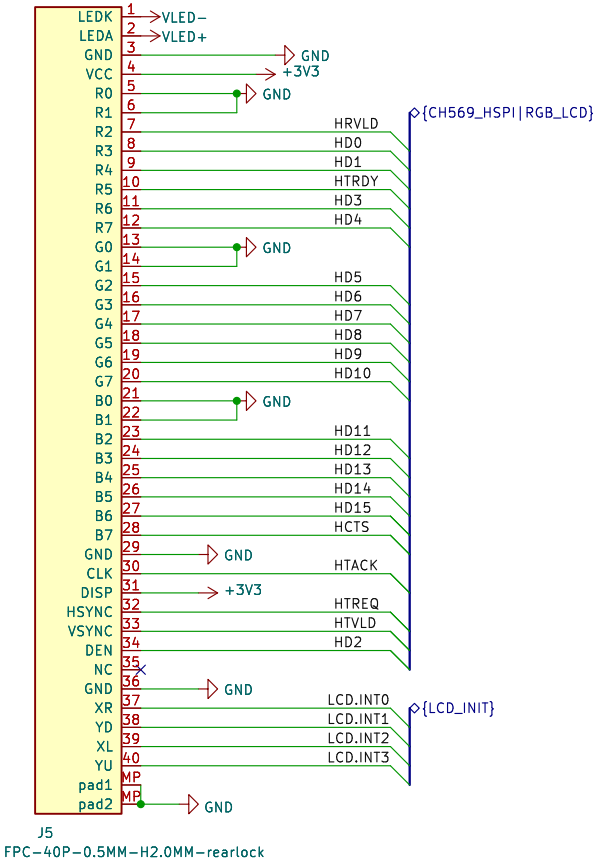
Sheet: /RGMII_ETHNET/
File: RGMII_sch.kicad_sch

Title: Tang_Mega_138K_Dock(NEO)

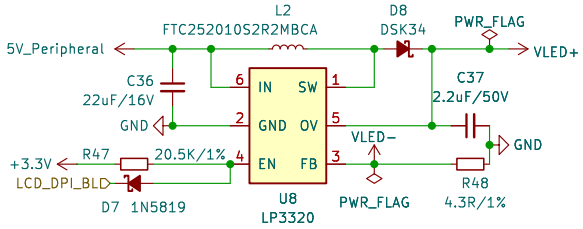
Size: A4	Date: 2025-03-30	Rev: 1.4
KiCad E.D.A. 8.0.6		Id: 11/21

RGB LCD Display

RGB LCD Display Connector

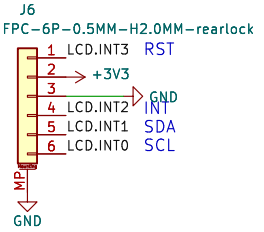


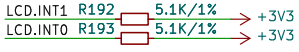
BACKLIGHT Driver



Feedback Voltage = 250mV
BackLight Current = 250mV/4.3R ≈ 58.1mA
7inch LCD QTM070TFT40-001 : IF=20*3=60mA

Capacitive Touch Connector





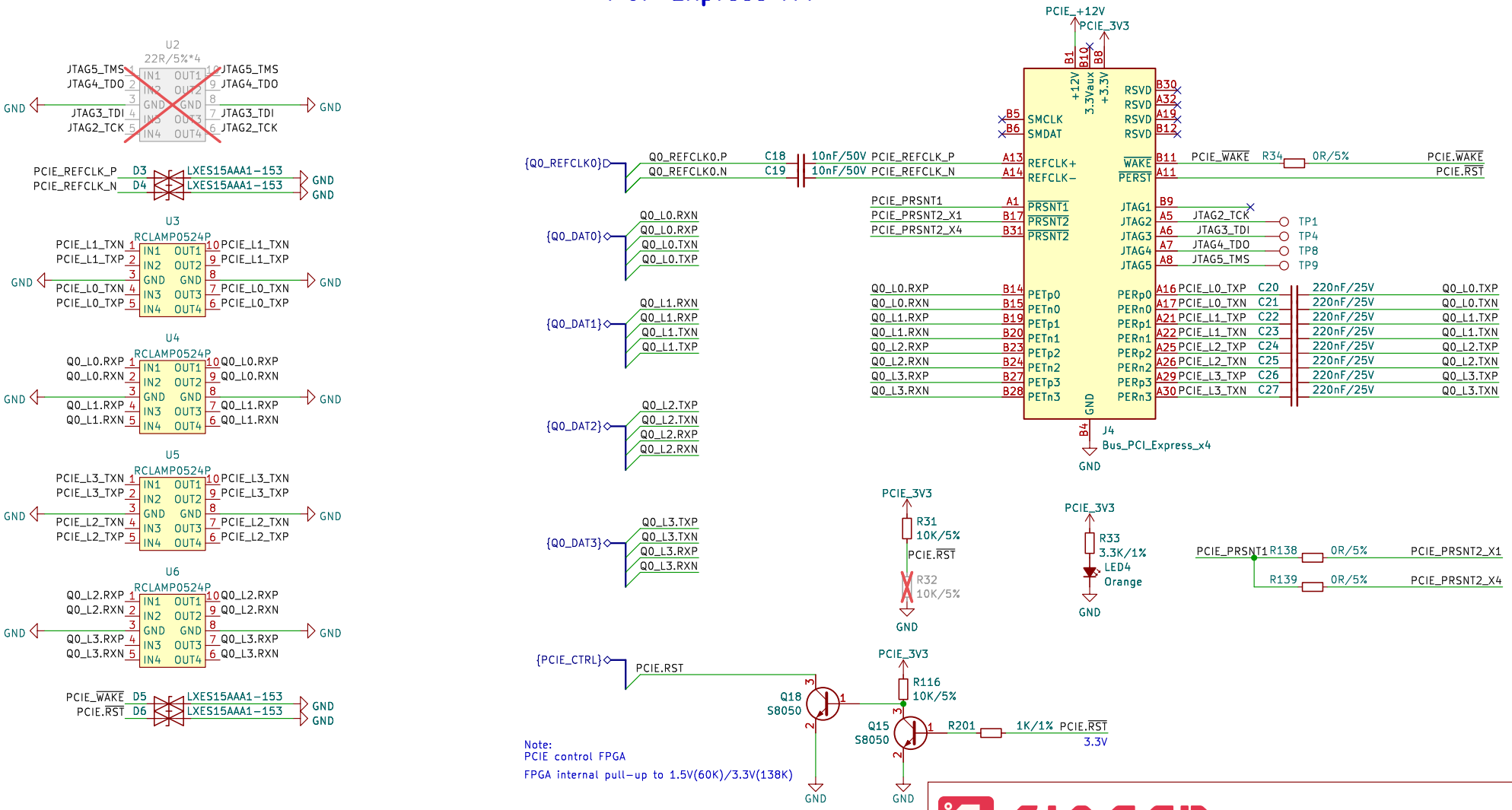
SiPEED

Sheet: /SYS_DPI/
File: LCD_DPI.kicad_sch

Title: Tang_Mega_138K_Dock(NEO)

Size: A4	Date: 2025-03-30	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 12/21

PCI-Express X4

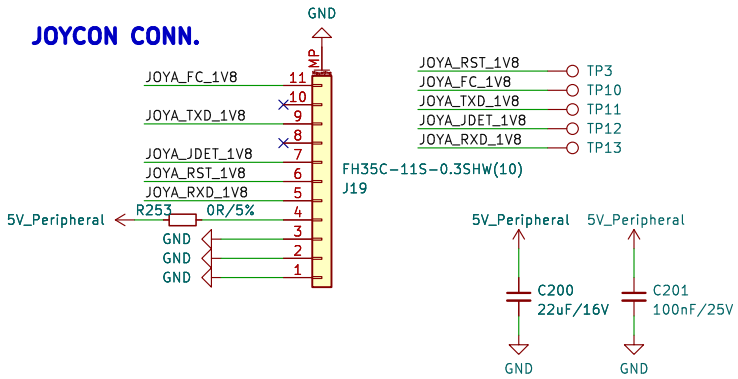


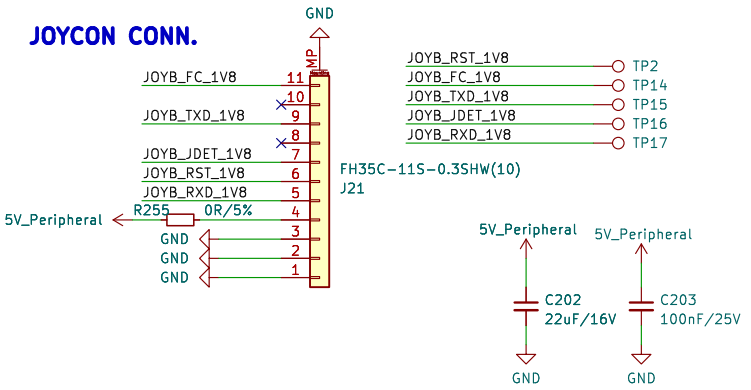
SiPEED

Sheet: /PCIE_GT/
File: SERDES_PCIE.kicad_sch

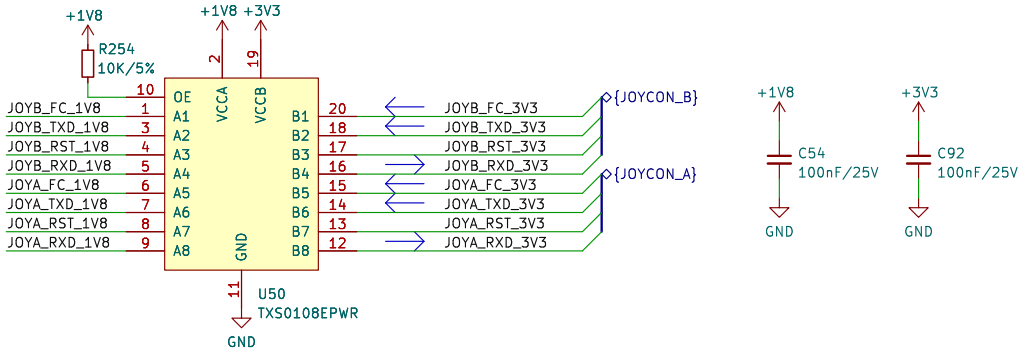
Title: Tang_Mega_138K_Dock(NEO)

Size: A4	Date: 2025-03-30	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 13/21





Bi-directional Voltage Translator

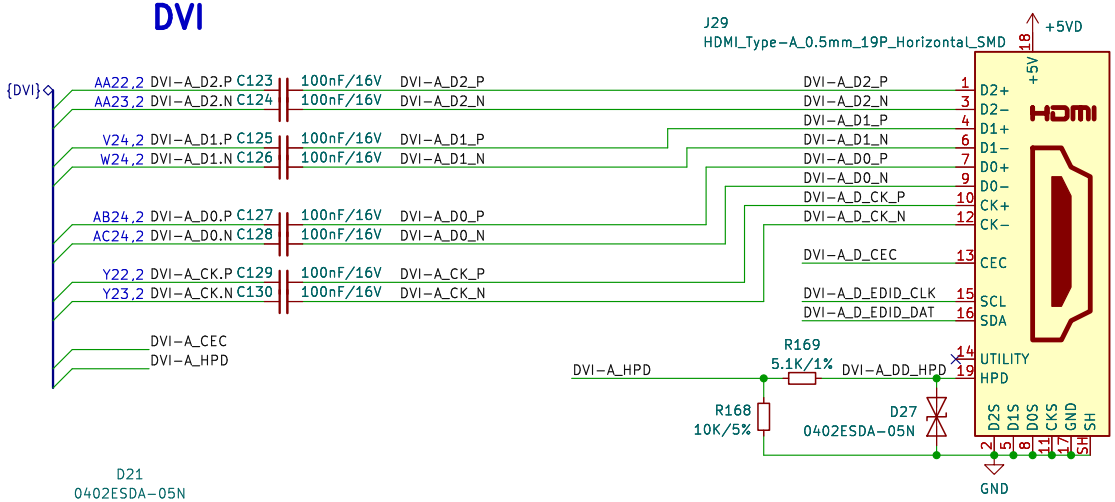


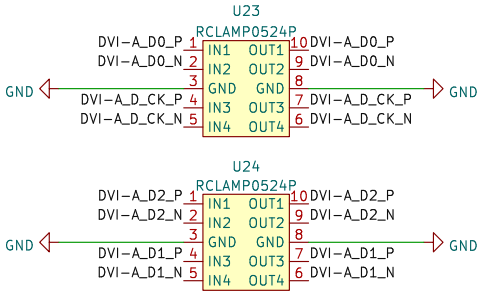
Note:
RST=0: Enable JOYCON
RST=1: Disable JOYCON

JDET=1:
HIGH when connected to console via Bluetooth.
Joy-Con will not send serial data post-handshake unless pin is pulled LOW by console.

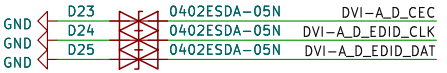
Sheet: /JOYCON/		
File: JOYCON.kicad_sch		
Title: Tang_Mega_138K_Dock(NEO)		
Size: A4	Date: 2025-03-30	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 14/21

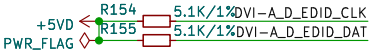
DVI

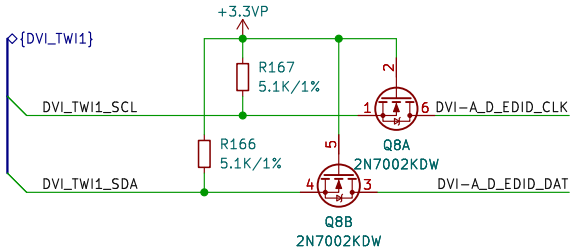


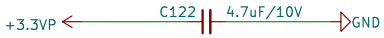






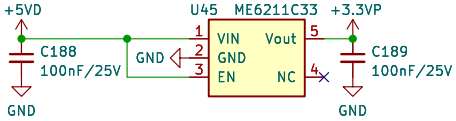


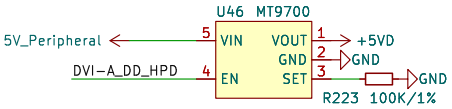




DVI Power control

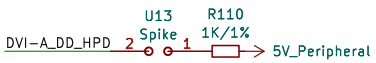
(Save Battery Power)

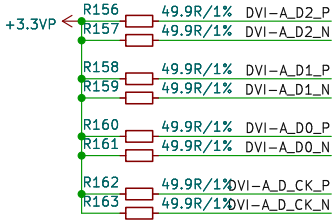


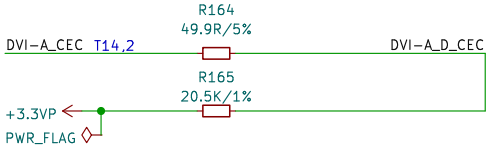


Insert HDMI : HPD=1
Unplug HDMI : HPD=0

Current limit: 68mA







SiPEED

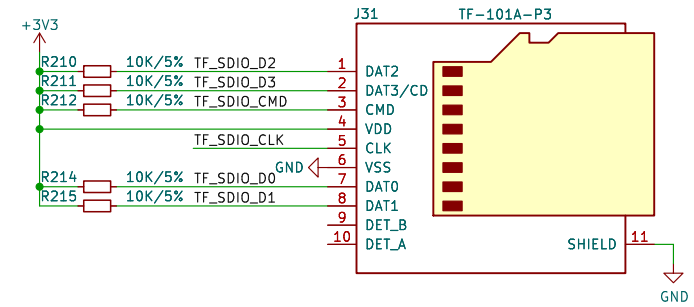
Sheet: /FPGA_TMDS_A/
File: DVI_A.kicad_sch

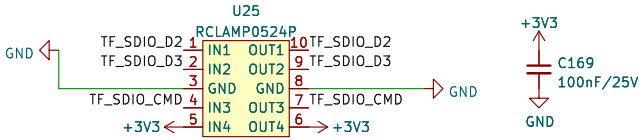
Title: Tang_Mega_138K_Dock(NEO)

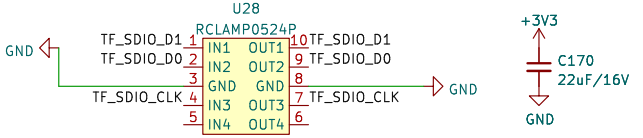
Size: A4	Date: 2025-03-30	Rev: 1.4
KiCad E.D.A. 8.0.6		Id: 15/21

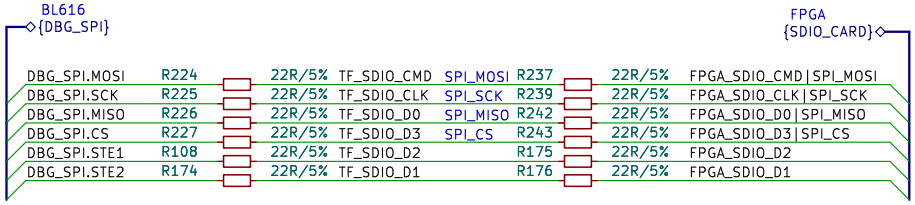
STORAGE

BL616 MICRO SD CARD





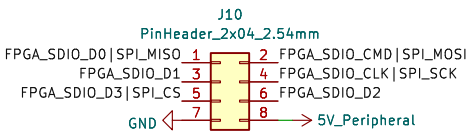




EVB mode : BL616_SPI high-Z , FPGA can work in SPI mode or SDIO mode

Game mode : BL616_SPI Master , FPGA_SPI Slave , FPGA_SPI_CS diasble , FPGA_SDIO_D1/D2 work as SPI_CS(input) or other function

Q1 : How to switch between EVB mode and Game mode?
A1: Switch DBG_ModeSel through SW2(6P Switch DIP)



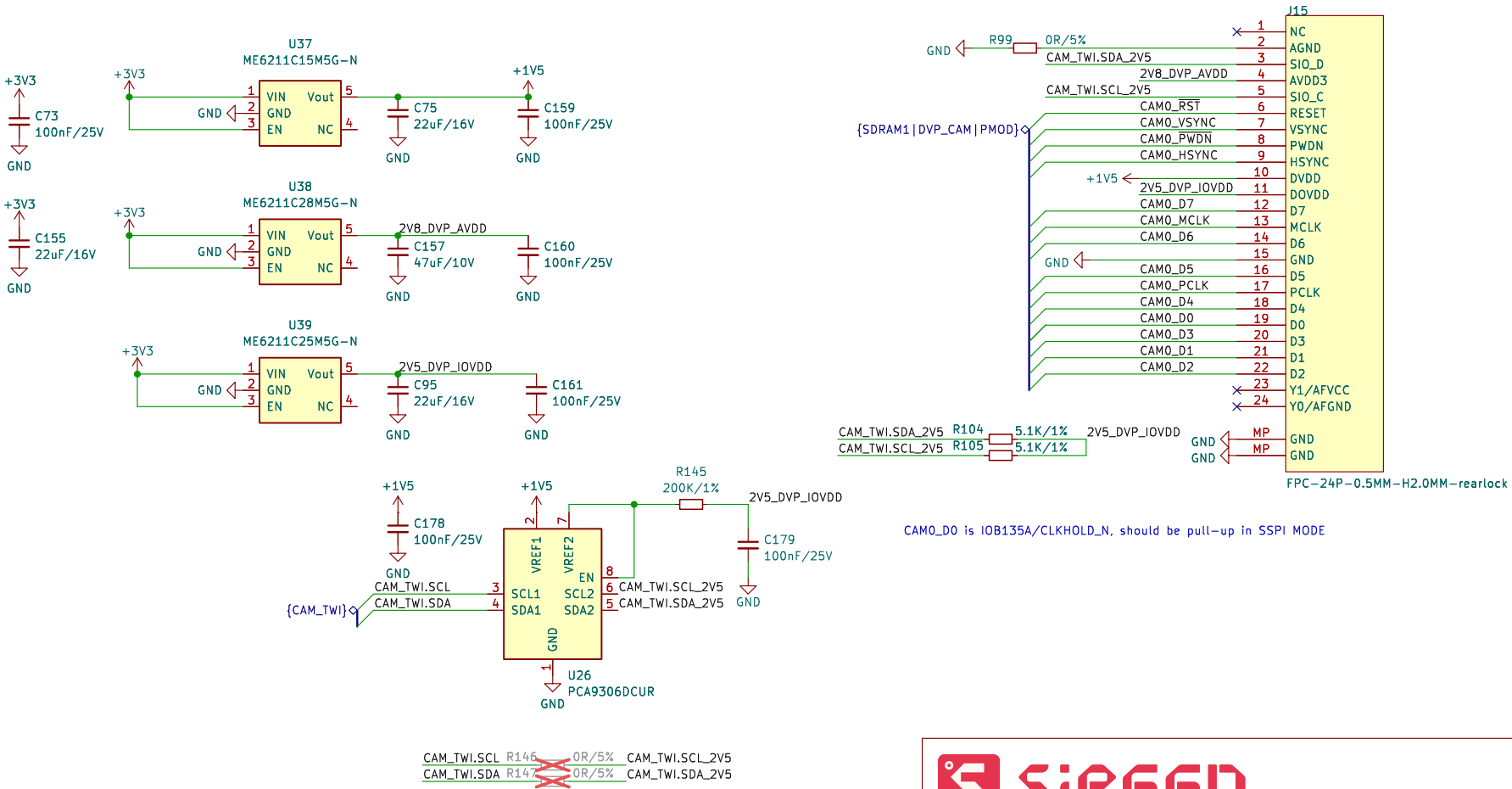
SiPEED

Sheet: /SYS_STORAGE/
File: SDIO&M.2.kicad_sch

Title: Tang_Mega_138K_Dock(NEO)

Size: A4	Date: 2025-03-30	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 16/21

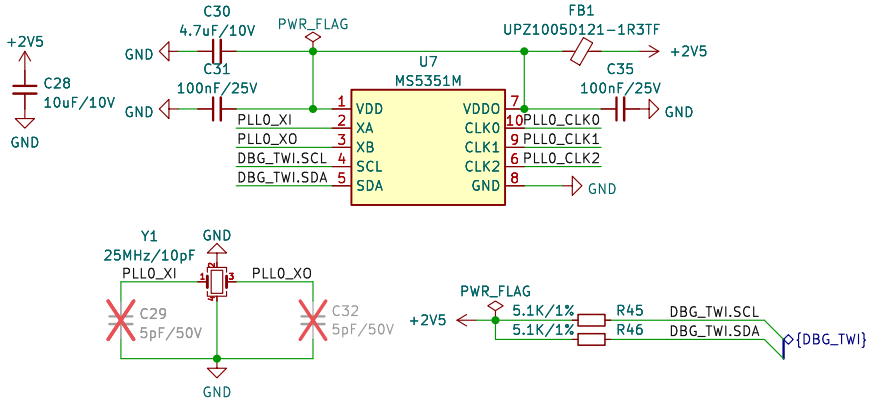
DVP-CAMERA

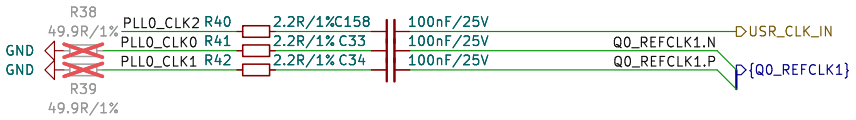


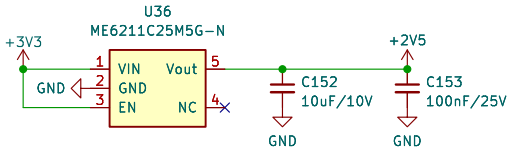
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File: DVP_CAM.kicad_sch		
Title: Tang_Mega_138K_Dock(NEO)		
Size: A4	Date: 2025-03-30	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 17/21

CLOCK

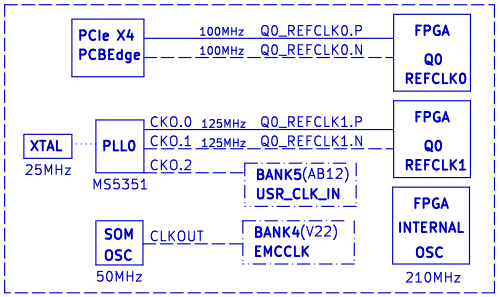
EXT. PLL







COLCK TREE



FPGA internal clock generator OSC.
The user can change the value of the option "Frequency Divider" to generate a different output clock frequency, which can be one of 64 values: 1/2, 1/3, 1/4, 1/6, 1/8, 1/10, ... or 1/126 of the oscillator frequency.

SiPEED

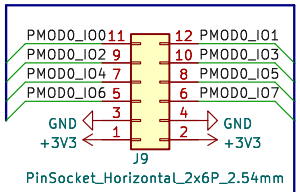
Sheet: /SYS_CLOCK/
File: CLOCK.kicad_sch

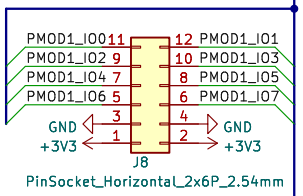
Title: Tang_Mega_138K_Dock(NEO)

Size: A4	Date: 2025-03-30	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 18/21

EX CONN.

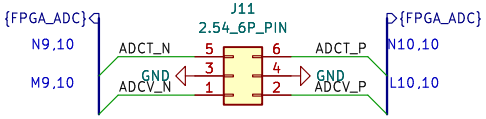
PMOD



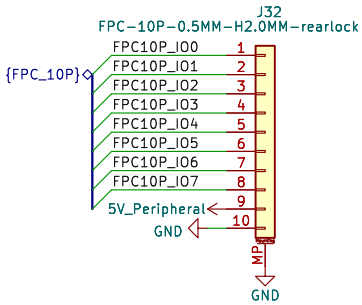


{SDRAM1|DVP_CAM|PMOD}

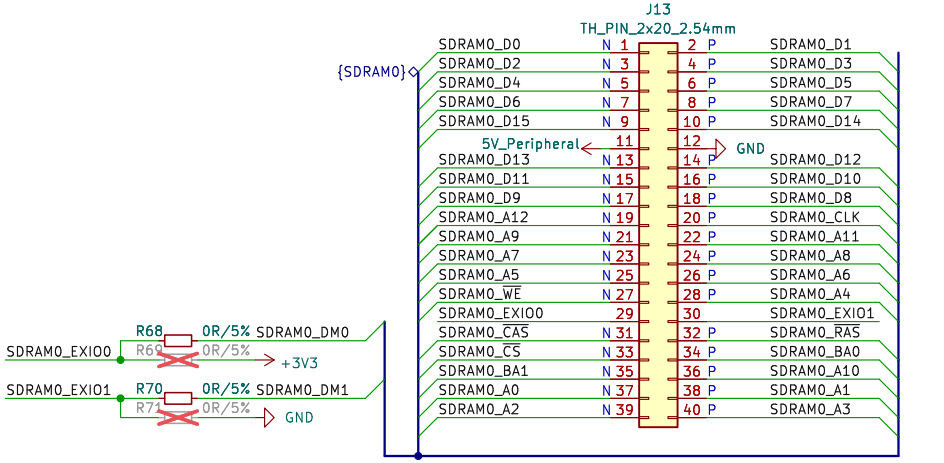
ADC CONN.



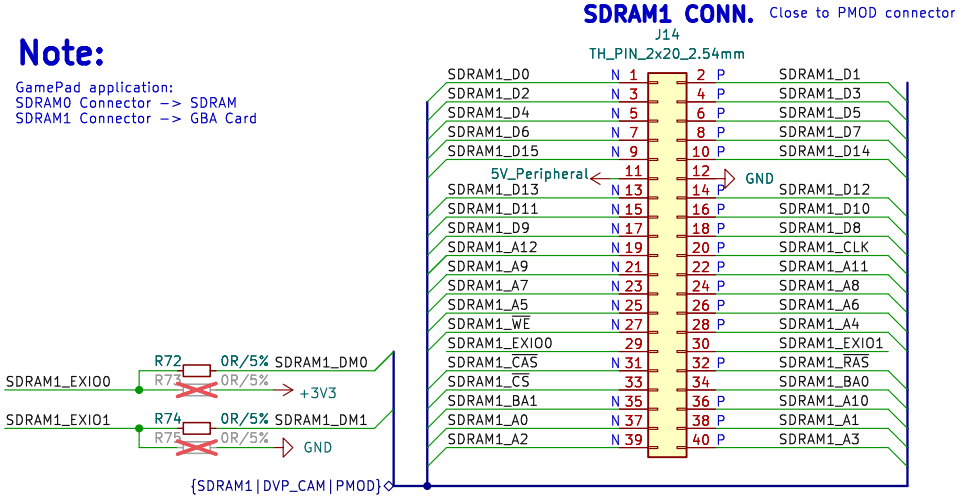
FPC10P CONN.



SDRAM0 CONN. Close to PCIE connector



SDRAM1 CONN. Close to PMOD connector



Note:

GamePad application:
SDRAM0 Connector -> SDRAM
SDRAM1 Connector -> GBA Card

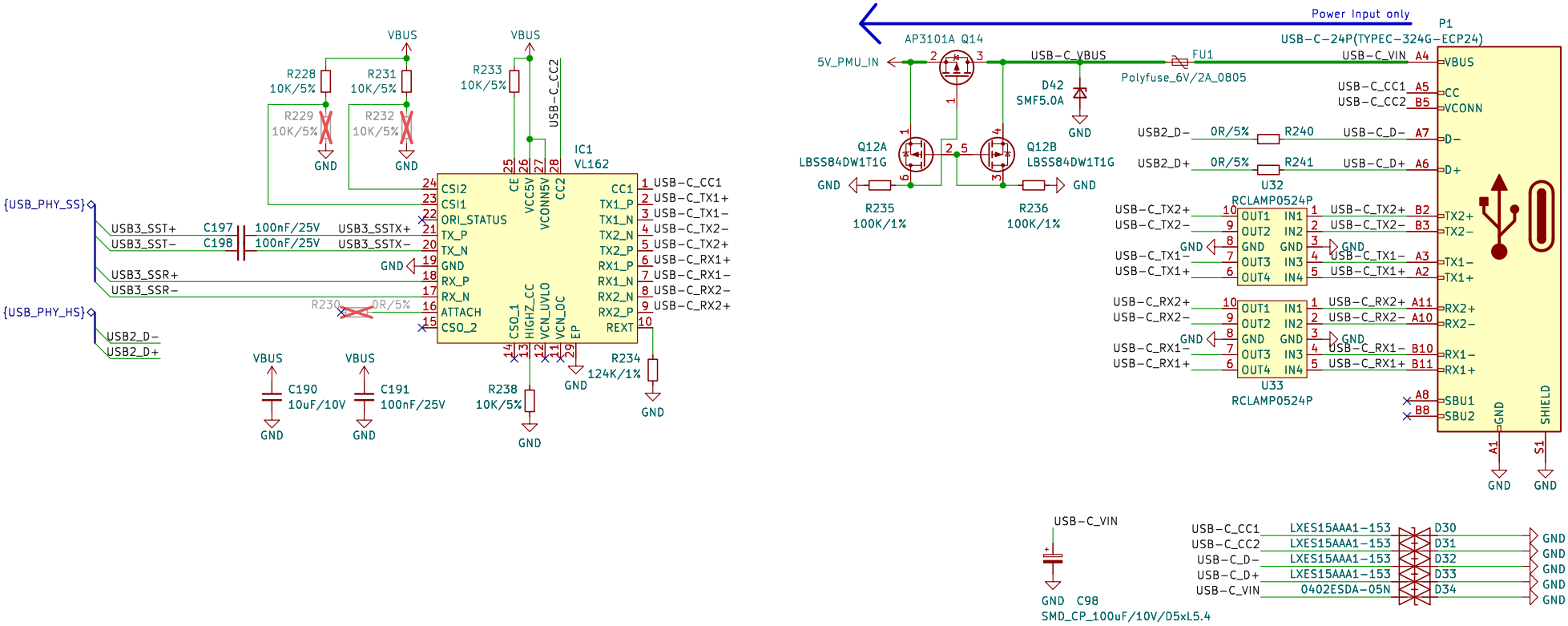
SiPEED

Sheet: /FPGA_EXT_CONN./
File: EX_CONN..kicad_sch

Title: Tang_Mega_138K_Dock(NEO)

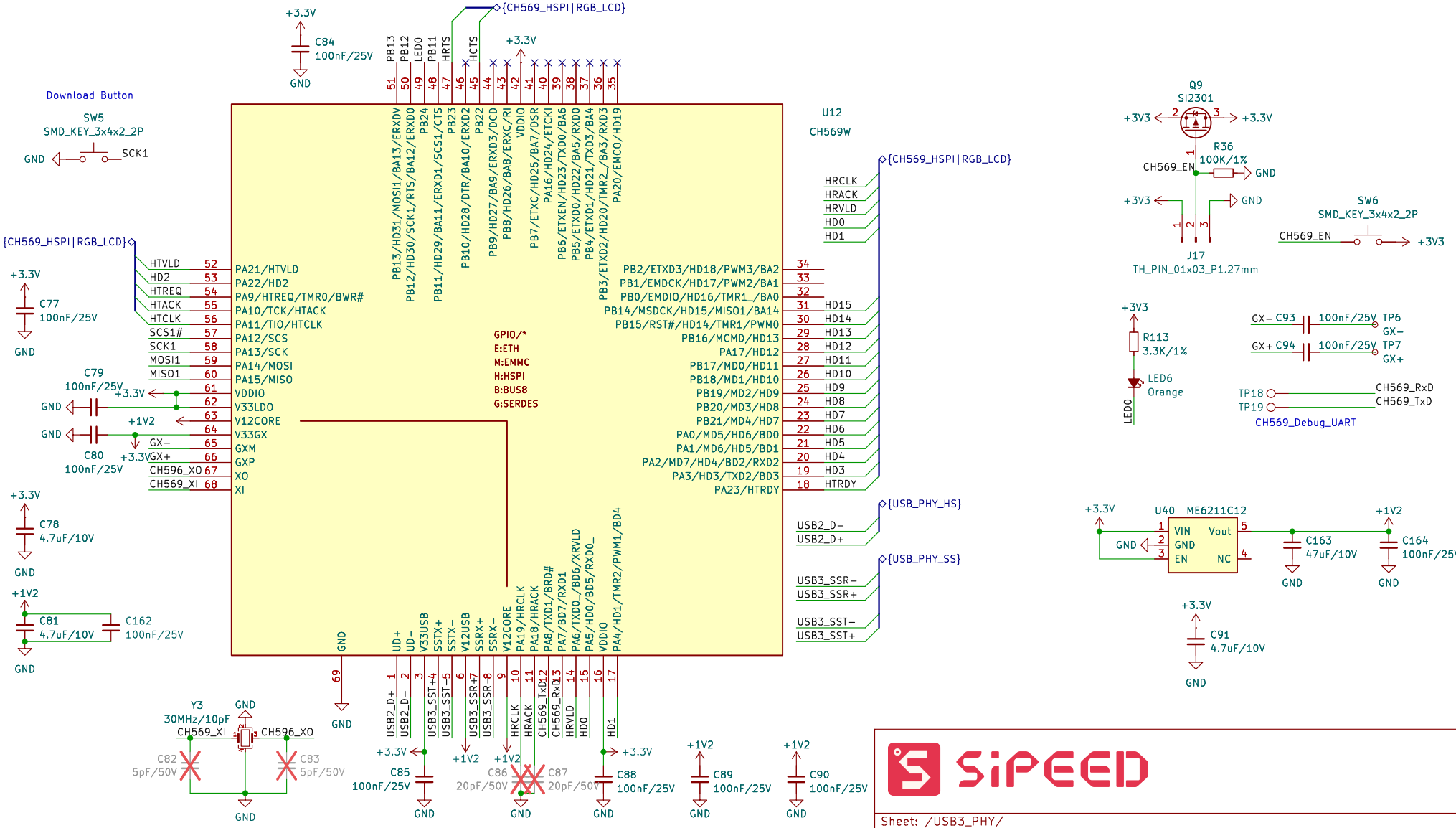
Size: A4	Date: 2025-03-30	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 19/21

CH569 ⟷ USB 2:1 Data Switch ⟷ CH569 USB-C Connector



Sheet: /SYS_USB-C_UFP/		
File: USBC_UFP.kicad_sch		
Title: Tang_Mega_138K_Dock(NEO)		
Size: A4	Date: 2025-03-30	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 20/21

USB3-PHY



Sheet: /USB3_PHY/		
File: USB3_PHY.kicad_sch		
Title: Tang_Mega_138K_Dock(NEO)		
Size: A4	Date: 2025-03-30	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 21/21